

Module 2 Reflective Journal

Student: Yuanyuan Kou

Group3

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1. Beginning the Lab

Before this lab, I did not have a clear understanding of what a notebook was. I first saw the .ipynb file as an unfamiliar type of document and was unsure where to begin. After opening it in Google Colab, I realized that a notebook is like an interactive workbook that combines explanations, Python code, results, and graphs. This understanding made the assignment feel more manageable because I could work through one cell at a time instead of treating the whole file as one large program.

2. Understanding the Roles of the Tools

The most important insight for me was seeing how pandas, NumPy, and Matplotlib have different but connected roles. Pandas organized the Iris data into a DataFrame that looked familiar because of my experience with Excel. However, I could see that a DataFrame is more powerful because Python can select, group, count, and summarize data automatically. NumPy handled the numerical calculations, such as the mean and standard deviation of sepal length. Matplotlib then changed numerical information into visual information through a scatter plot and a bar chart. I began to understand that data analysis is a workflow: first organize the data, then calculate useful information, and finally visualize patterns.

3. Learning from the Iris Dataset

The visualizations helped me understand why data exploration is necessary before building a machine learning model. The bar chart showed that the dataset is balanced because all three flower species have 50 samples. In the scatter plot, setosa appeared more separate, while versicolor and virginica overlapped when sepal length and sepal width were compared. This made me question whether petal measurements would separate the species more clearly. I also learned that features are the measurements used for prediction and that the species column represents the target class that a future model would try to predict.

4. Challenges and Problem-Solving

One challenge was understanding why the cells must be run in order. At first, clicking a Run button seemed like a simple action, but I learned that later cells depend on variables created earlier. For example, the DataFrame named df must exist before it can be grouped, summarized, or plotted. This helped me see that an error message may not mean the code itself is wrong; it may mean that

an earlier step was skipped. Running all cells from the beginning became an important way to test whether the notebook was reproducible.

5. How My Thinking Changed

This lab changed my view of machine learning because I had expected the main task to be training a model. Instead, I learned that understanding and preparing data comes first. A model can only learn from the information it receives, so checking the structure, balance, measurements, and visual patterns of a dataset is an essential part of the process. I still need more practice reading Python syntax, but I now understand the purpose of the main tools and feel more confident approaching a notebook systematically.

6. Next Steps

My next goal is to become more comfortable changing small parts of the code and predicting what the output will be. I would like to compare different pairs of Iris features and learn which features are most useful for classification. I also want to practice reading error messages instead of feeling discouraged by them. This lab gave me a foundation for the next module, where the data exploration process will connect to an actual machine learning classifier.